

PROBLEM STATEMENT ID : 26168

PROBLEM STATEMENT TITLE :

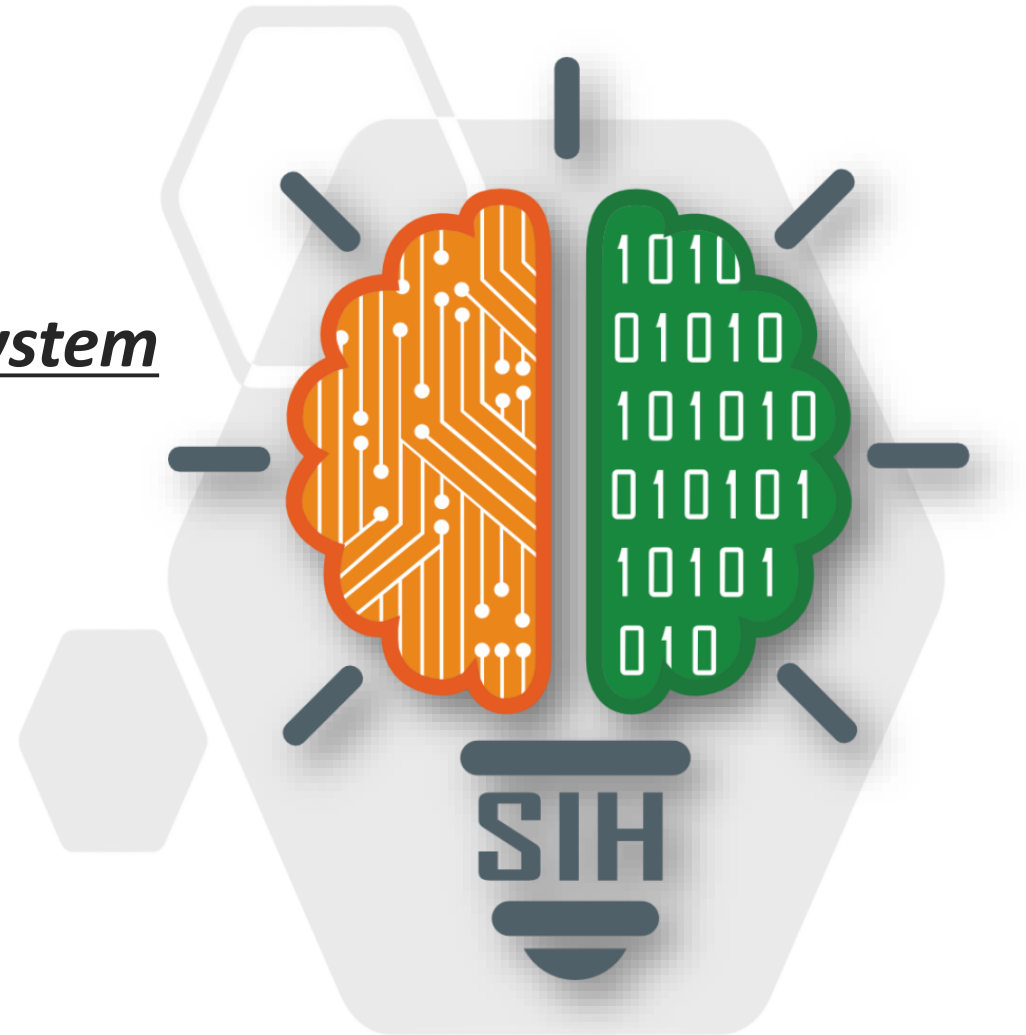
**AI-ML based Intelligent Dead Reckoning System
for seamless navigation**

THEME – SMART VEHICLES

PS CATEGORY - SOFTWARE

TEAM ID - 26168

TEAM NAME – OFFGRID



The problem

- GPS loss interrupts navigation in tunnels and dense streets.
- Unaided phone sensors accumulate position error.

What's different

- No vehicle connection or wheel sensors.
- Learned uncertainty controls correction strength.
- Models run locally on the phone.

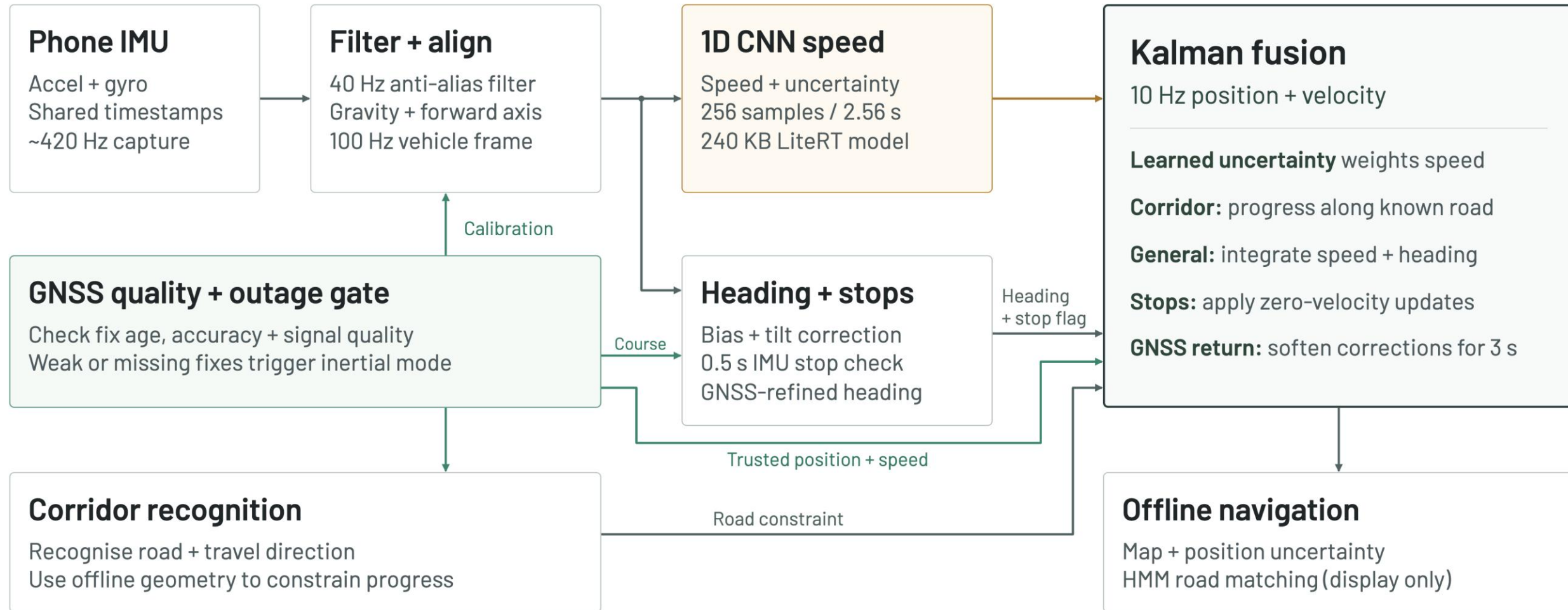
Our solution

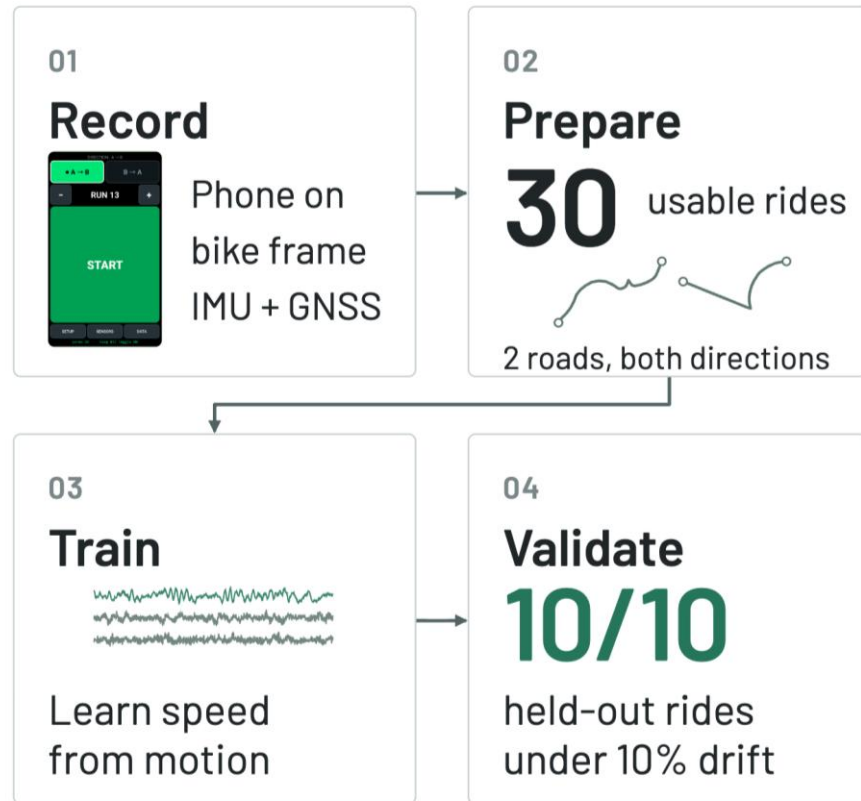
Navigation through GPS outages using the phone's own motion

- Auto-calibration aligns the phone to the vehicle.
- AI estimates speed from phone motion.
- Fusion combines speed, heading and offline roads.
- Stop detection limits drift while stationary.
- GPS corrections blend back when signals return.

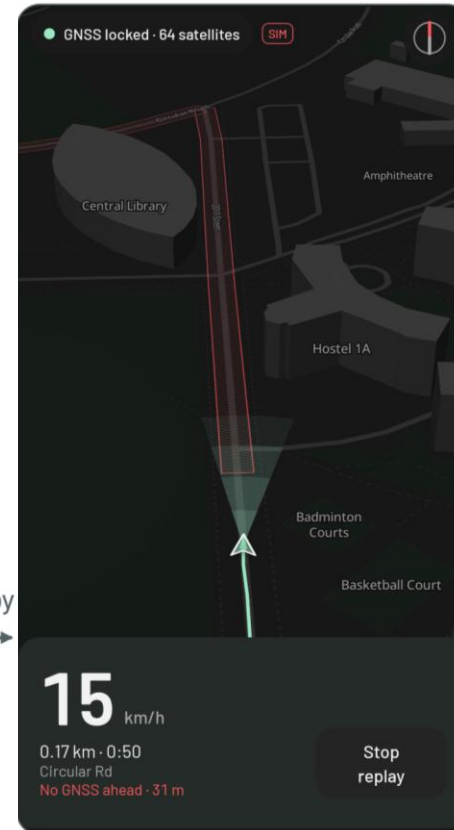
10 Hz position updates

Offline navigation

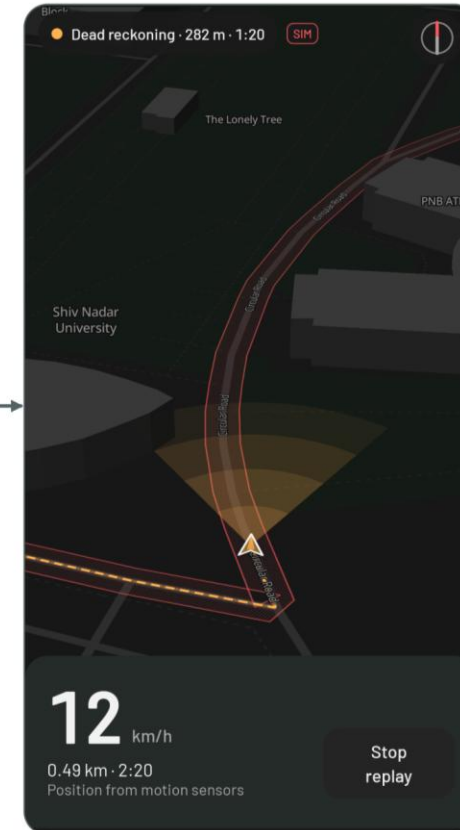




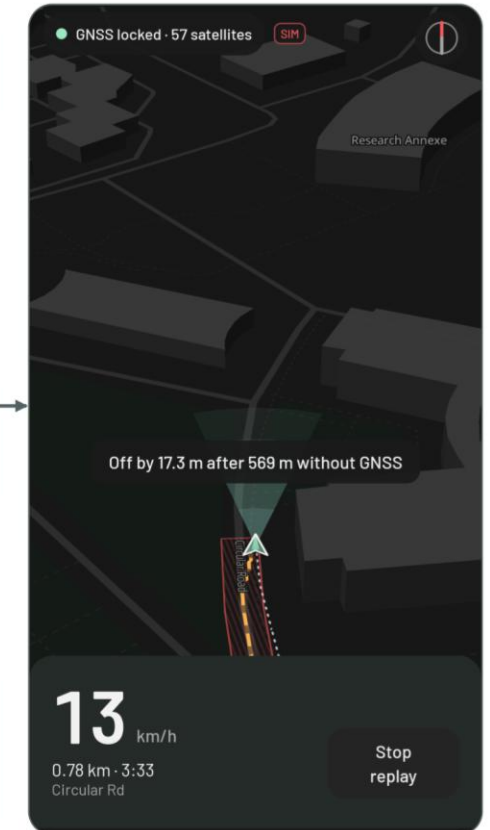
GPS locked



Dead reckoning



GPS returns



[WATCH OUR WORKING PROTOTYPE!](#)

17.3 m error after 569 m without GPS

Replay demonstration. GPS recorded for comparison only.

Feasibility and viability

- **Runs on the phone**
240 KB model with 10 Hz position updates.
- **Corridor replay evidence**
10/10 held-out route-2 rides below 10% drift.
- **Works offline**
Local inference and bundled road maps.

Next steps

- **Broader vehicle coverage**
Train across more vehicles and phone models.
- **Longer GPS outages**
Refine calibration and road constraints.

Impact and benefits

Potential benefits

- **Navigation continuity**
Position tracking through GPS blackouts.
- **Delivery reliability**
Fewer missed turns and avoidable delays.
- **Emergency response**
Position awareness where GPS is blocked.
- **Lower adoption cost**
Existing phones, without vehicle wiring.
- **More efficient trips**
Fewer detours could cut fuel use and emissions.

Learned motion

Research informing the speed model

01 **Bicycle inertial odometry (MoE)**

Qiao et al., 2025/26 preprint

Learned velocity for bicycle motion.

02 **DeepOdo**

Wang et al., IEEE TIM, 2023

Smartphone IMU to forward speed.

03 **DVSE**

Xiao et al., 2025 preprint

GNSS supervision and phone alignment.

04 **OdoNet**

Tang et al., 2021 preprint

CNN speed aiding without wheel sensors.

05 **AI-IMU Dead-Reckoning**

Brossard et al., 2019 preprint

Learned noise adaptation in fusion.

06 **TLIO**

Liu et al., IEEE RA-L, 2020

Learned uncertainty as a filter input.

Fusion and evaluation

Methods, constraints and benchmark

07 **IO-VNBD**

Onyekpe et al., Data in Brief, 2021

External vehicle benchmark dataset.

08 **KF-GINS**

i2Nav, Wuhan University

GNSS/INS error-state filter reference.

09 **Wheel-INS**

Niu et al., IEEE TVT, 2021

Vehicle constraints and calibration.

10 **Zero-velocity detection**

Wagstaff et al., 2019 preprint

Foot-mounted stop-detection methods.

11 **Hidden Markov map matching**

Newson & Krumm, ACM GIS, 2009

Probabilistic matching to road networks.

12 **Route-constrained fusion**

Cui et al., 2026 preprint

Known-route geometry to limit drift.

Software and map data

Tools and data used in the prototype

13 **PyTorch / TensorFlow-Keras**

Training and model conversion

CNN training and matching model export.

14 **Android / Kotlin / Compose**

Sensor logger and navigation app

IMU and GNSS capture, services and UI.

15 **Google LiteRT**

Local model execution

Android inference for the speed model.

16 **MapLibre Native**

Map rendering

Offline map display and camera control.

17 **OpenStreetMap / Protomaps**

Road data and offline basemap

Campus roads and bundled PMTiles.

18 **Microsoft footprints / Esri**

Buildings and satellite view

Buildings and online satellite imagery.